First Best Implementation with Costly Information Acquisition*

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Abstract

We study a mechanism design model with flexible but costly information acquisition. There is a principal and $I \geq 4$ agents. The principal and the agents share a common prior over the set of payoff-relevant states of the world. The principal proposes a mechanism to the agents who can then acquire information about the state of the world by privately designing a signal device. As long as it is costless for each agent to acquire a signal that is pairwise independent from the state of the world, we show that there exists a mechanism which allows the principal to implement any social choice rule at zero information acquisition cost to the agents.

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1 Introduction

In most mechanism design problems, there is a collection of agents who have exogenously given private information, and there is a principal who desires to implement a social choice rule, by designing a mechanism which incentivizes the agents to reveal their information.

In many problems in practice, however, the agents' private information is often a consequence of their own (possibly costly) *information acquisition*. For example, bidders in an oil-tract auction (Wilson, 1969) may conduct test drills; bidders in a spectrum auction may conduct market research; voters in a presidential election may investigate the candidates' past political activities; members of a hiring committee may study the job applicant's background in order to see whether he is fit for the job.

Importantly, in such situations, a mechanism in place does not only affect each agent's incentive of reporting the acquired information truthfully, but also affects his choice of *what kind of information* to acquire. In this sense, the properties of desirable mechanisms could potentially be very different from those which only guarantee truth-telling incentives for a given information structure.

Although this issue is already relevant in single-agent environments,[1] the degree of complexity is even higher in a multi-agent environment: in principle, flexibility of each agent's information acquisition action does not only mean flexibility in terms of his signal's informativeness about the payoff-relevant state (e.g., the amount of oil in a tract), but also means flexibility in terms of his opponents' signals. This issue of higher-order information / beliefs is unique in multi-agent environments.[2] As far as we understand, when it comes to higher-order information, there is no obvious or universally agreed upon class of information acquisition cost functions.[3] Hence, in this paper, we take an agnostic approach by essentially assuming that an agent's information cost only depends on its informativeness about the payoff-relevant state, but not on the others' signals. We believe that it is a reasonable "first-step" assumption, but we must admit that our

[1] Mensch (2020) studies a mechanism design problem with a single agent. See also Section 1.1.

[2] Of course, this issue can be avoided if we impose some *ad hoc* assumptions such as conditional independence, although that is not the scope of the paper.

[3] See Section 1.1 and 5.1.

result crucially depends on this assumption. We discuss it in Section 5.1.

We consider a model with four or more agents, and assume that it is costless for each agent to acquire a signal that is pairwise independent from the state of the world.[4] The principal and agents share a common prior, and none of them has any private information at the beginning. We show that there exists a mechanism which allows the principal to implement any social choice rule at zero information acquisition cost to the agents. The mechanism recommends each agent to choose a special information acquisition action, which satisfies the *individually-uninformative-but-aggregately-revealing* property of Zhu (2021) (and each agent finds it optimal to obey this recommended action). The *individually-uninformative* part means that each agent's signal on its own is pairwise independent from the payoff-relevant state, which guarantees that his information cost is zero. The *aggregately-revealing* part means that the principal, by observing all the agents' reports — in fact, any two of them, — can correctly identify the true payoff-relevant state. The fact that only two is enough, together with the fact that there are four or more agents, enables the principal to detect any unilateral deviation. It establishes the incentive compatibility of the mechanism.

1.1 Related Literature

In the literature of information acquisition in mechanism design, we usually consider restricted / less flexible spaces of information (see, for example, Bergemann and Välimäki (2002) for efficient mechanism design, Shi (2012) and Bikhchandani and Obara (2017) for optimal auction design, and Persico (2004), Gerardi and Yariv (2008), Gershkov and Szentes (2009), and Zhao (2016) for committee design with information acquisition.[5])

Mensch (2020) studies mechanism design with a single agent's flexible and costly information acquisition, building on the rational inattention framework (Sims (2003))[6]. Flexible information

[4]The literature on cost of information proposes and discusses a variety of possible cost functions (see Section 1.1), but it seems to be universally accepted that uninformative signals about the state of the world are costless.

[5]Restricting to the class of conservative rules, Li (2001) solves for the optimal degree of conservatism in committee design. The optimally chosen conservative rule outperforms the *ex post* optimal rule.

[6]Mensch (2020) also considers a multiple-agent extension of his model, but restricts attention to symmetric mechanisms in an independent private values setting, in which agents can acquire information about their own

acquisition is also considered by Roesler and Szentes (2017) in the context of buyer-optimal information in monopoly pricing,[7] by Bergemann et al. (2017) and Brooks and Du (2021) in the context of seller-pessimal information in common-value auction, and by Yamashita (2018) in private-value auction. All these papers have a common feature that there is a single entity, "nature", who chooses the information structure (of one or multiple agents). In contrast, in our model, each of multiple agents acquires information in a decentralized manner, leading to a very different conclusion.

The information structure we employ was proposed in the context of mechanism design by Zhu (2021), who studies information disclosure by a mechanism designer. It builds on the idea of the one-time pad, unbreakable encryption (Shannon, 1949).[8]

This key information structure makes the agents' acquired information statistically dependent. In quasi-linear environments, Crémer and McLean (1988) show that the principal can extract full surplus from the agents. Although the extreme positivity of the results is a common feature of our paper and theirs, the two problems are quite different. First, our paper does not assume quasi-linearity. Second, their side-bet mechanism exploits an exogenously given correlated signal structure, and it is not clear if it can be induced as an equilibrium information structure given some reasonable spaces of information acquisition actions.[9] In our case, the above mentioned information structure is indeed an equilibrium outcome, even though each agent can potentially acquire information independent from the others' signals.

In non-quasi-linear environments, such as collective decision-making in committees, the first best outcome is generally not implementable under the commonly imposed restrictions on information acquisition technologies. For example, Li (2001) and Persico (2004), assuming that

values, but cannot acquire any information about others' values.

[7]See also Condorelli and Szentes (2020), though they also consider non-information changes of the agent's private information distribution.

[8]See also Kalai et al. (2010); Renou and Tomala (2012); Renault et al. (2014), in the context of games of communication network. Peters and Troncoso-Valverde (2013) apply this idea in mechanism-design games with multiple principals, and Liu (2015) applies it in his concept of individually uninformative correlating device. Our construction is most directly related to Zhu (2021).

[9]Bikhchandani (2010) shows that, indeed, an agent in the Crémer-McLean mechanism may have a strong incentive of acquiring information about others.

the agents have access to conditionally independent signals, show that the first best outcome is not attainable. In contrast to the previous results, we show that correlated information acquisition resolves the free-riding issue in collective decision-making in committees, which, in turn, helps to implement the first best outcome.

There is a growing literature on the cost of flexible information in decision environments. See for example Sims (2003), Matejka and Mckay (2015), Caplin and Dean (2015), and Pomatto et al. (2020). Usually the main focus is on the cost of acquiring more / less precise information about the payoff-relevant state, and its relationship with a single decision maker's optimal choice. The framework, however, has been applied in multi-player problem (i.e., in games), such as in coordination games (Yang (2015); Morris and Yang (2021); Denti (2020). In particular, Denti (2020) proposes a model of unrestricted information acquisition in games in which, as in our paper, the players can endogenously learn about a payoff-relevant state and actions of other players.

2 Model

2.1 Setup

There exist a principal and $I \geq 4$ agents, and a finite set of payoff-relevant states Θ. Each agent i's payoff is denoted $u_i(d, \theta)$, when a social decision $d \in D$ is selected in state θ.[10] For example, in an auction, d may comprise a vector of each bidder i's winning probability and his expected payment, and u_i is quasi-linear in the payment part. Later, his payoff *net* his information acquisition cost is considered as his objective.

At the beginning, neither the principal nor any of the agents know θ, but they share a common prior μ_0 on Θ. In what follows, it is useful to imagine that there exists an unobservable fundamental state of nature, $x \in X$, where X is a measurable space associated with a probability measure $\mathbb{P}$. We assume $X = [0, 1]$ with a Borel σ-algebra and x is uniformly distributed on $[0, 1]$.[11] Taking a richer space of the fundamental state of nature does not change our

[10] We can endow the principal with his own payoff function $u_0(q, \theta)$, though it is not necessary.

[11] See Gentzkow and Kamenica (2017) who adapt this approach in the context of multi-sender Bayesian

results. Then, we define θ as a measurable function of x, $\theta = \theta(x)$. $\theta \sim \mu_0$ means that $\int_0^1 1_{\{\theta(x)=\theta\}} dx = \mu_0(\theta)$ for each $\theta \in \Theta$.

The agents can acquire *costly* information about θ by generating private and possibly correlated signals, whereas the principal cannot. More specifically, let S_i be the set of possible signals agent i may potentially obtain by any information acquisition technology. Potentially, S_i (in particular, its size) may be a part of i's choice, but assuming exogenous S_i simplifies our analysis. Besides, as long as $|S_i| \geq |\Theta|$, the same results obtain. i's information acquisition action is a measurable function $\sigma_i : [0,1] \to S_i$, such that, once x (and hence $\theta = \theta(x)$) is realized, then i observes $s_i = \sigma_i(x)$. Let Σ_i denote the set of all such measurable functions, defining i's information acquisition action space. We assume that information acquisition is fully private in the sense that neither the principal nor any other agent observes which action i takes and which signal is realized. Agent i's objective is the *net* payoff $u_i(d, \theta) - c_i(\sigma_i)$. We assume the following property of the cost function:

Assumption 1. *$c_i(\sigma_i) \geq 0$ for any $\sigma_i \in \Sigma_i$. $c_i(\sigma_i) = 0$ if its induced s_i and θ are independently distributed, that is, for each $s_i \in S_i$ and $\theta \in \Theta$:*

$$\int_0^1 1_{\{\sigma_i(x)=s_i \text{ and } \theta(x)=\theta\}} dx = \left(\int_0^1 1_{\{\sigma_i(x)=s_i\}} dx\right)\left(\int_0^1 1_{\{\theta(x)=\theta\}} dx\right)$$

It is a quite general specification, which includes many popular cost functions as special cases: entropy-based cost functions, more general posterior-separable cost functions, and even some posterior-non-separable ones.[12] However, it should be noted that we do not allow i's information acquisition cost to depend on others' information acquisition actions. In other words, the cost of information acquisition for each agent depends only on how much this agent individually learns about θ, and not on how much other agents learn. We think it is a reasonable choice, especially as a first-step approach, but it is a restriction, and our result hinges on it.[13]

persuasion.

[12]See the literature of cost of information, such as Sims (2003), Matejka and Mckay (2015), Caplin and Dean (2015), and Pomatto et al. (2020).

[13]For example, Strulovici (2021) considers a situation where some hard information is scarce; Denti (2020) considers a general interdependent information cost. As far as we know, there is no general consensus in the

Given any σ, a joint distribution for (θ, s) is determined, which we denote by $\alpha \in \Delta(\Theta \times S)$. If we want to make its dependence on σ more explicit, we denote it as α_σ.

2.2 Mechanism

The principal faces both hidden action and hidden information of each agent. The principal can commit to a mechanism at the ex ante stage in order to control the agents' incentives. More specifically, following the literature, we let the principal (i) send a message privately to each agent before his information acquisition action, and (ii) collect a message privately from each agent after observing a signal. Formally, a mechanism comprises $(R, \rho; M, \delta)$ where $R = (R_i)_{i=1}^I$ and $M = (M_i)_{i=1}^I$; R_i denotes the set of messages that the principal can send to each agent i; M_i denotes the set of messages that each agent i can send to the principal; $\rho \in \Delta(R)$ is a distribution over the principal's messages, and $\delta : R \times M \to D$ denotes the decision rule.

The timing of the game is summarized as follows:

$t = 0$: $x \sim U(0, 1)$ is drawn (though no one observes it).

$t = 1$: The principal designs a mechanism $(R, \rho; M, \delta)$.

$t = 2$: After observing the mechanism and receiving $r_i \in R_i$, each agent i privately chooses his information acquisition action $\sigma_i \in \Sigma_i$.

$t = 3$: Each agent i privately observes $s_i = \sigma_i(x)$, and privately sends $m_i \in M_i$ to the principal.

$t = 4$: The principal executes $d = \delta(r, m)$ where $m = (m_i)_{i=1}^I$.

Because no agent observes the other agents' actions or information (even noisily) at all, we consider Bayesian equilibrium as a solution concept. Then, applying the revelation principle of Forges (1986), we focus on *direct* mechanisms where (i) the principal directly recommends an information-acquisition action to each agent, and each agent directly reports a signal to the

literature which types of correlation structure should be more or less costly than others. Perhaps it depends on specific contexts of interest. See also our discussion in the concluding section.

principal, and (ii) each agent finds it optimal to obey the recommended action and truthfully report his signal.[14]

Formally, a direct mechanism comprises $((\sigma_i)_{i=1}^I, \delta)$, where the principal recommends $\sigma_i \in \Sigma_i$ privately to each agent i,[15] and executes $\delta(s) \in D$ if the agents report $s = (s_i)_{i=1}^I \in S$. It is a feasible direct mechanism if it satisfies the following constraints: for any $\sigma_i' \in \Sigma_i$ and $\tau_i : S_i \to S_i$,

$$\sum_{\theta, s_i, s_{-i}} (u_i(\delta(s_i, s_{-i}), \theta)\alpha_{\sigma_i, \sigma_{-i}}(\theta, s_i, s_{-i})) - c_i(\sigma_i) \geq \sum_{\theta, s_i, s_{-i}} (u_i(\delta(\tau_i(s_i), s_{-i}), \theta)\alpha_{\sigma_i', \sigma_{-i}}(\theta, s_i, s_{-i})) - c_i(\sigma_i').$$

That is, each i must find it optimal to obey the recommended σ_i and report the realized s_i truthfully.

Although the constraints are concisely summarized by the simple-looking inequalities above, they can potentially be very complicated. First, changing σ_i affects the joint distribution α of (θ, s) and the agent's cost, but the way it affects α is somewhat limited because i cannot affect $-i$'s information structure. In particular, the marginal distribution induced by α on (θ, s_{-i}) must not change at all. Second, an agent may potentially want to make a double deviation, that is, changing σ_i and at the same time changing his reporting strategy.

Remark 1. *Here, we do not explicitly impose each agent's individual rationality / participation constraint. It is not difficult to accommodate this constraint: let us require that any feasible direct mechanism must have an extra message ϕ_i (a "non-participation" message) so that i's message space is now $S_i \cup \{\phi_i\}$, and $\delta(\phi_i, m_{-i})$ is some specific allocation (a "non-participation allocation") for agent i, for any given m_{-i}. Accordingly, the individual rationality constraints,*

[14] The proof proceeds as follows. First, imagine an auxiliary game where there is no principal, but instead, there is a fictitious player ("player 0") who is indifferent across all decisions in any state. At first, each agent i plays σ_i privately, and then observes the realized signal s_i privately. Then, (without any communication), player 0 chooses $d \in D$. Interpreting this as a baseline extensive-form game, it is easy to see that our current game (with the principal) is the mediated communication game of this auxiliary game in the sense of Forges (1986) (see also Myerson (1986)). Thus, her revelation principle applies.

[15] Here, we focus on a deterministic recommendation of σ, rather than any stochastic recommendation. Accordingly, δ is denoted simply by $\delta(s)$ instead of $\delta(r, s)$. Our results do not change with stochastic recommendations.

both at the ex ante and ex interim stages are captured as a part of the above incentive compatibility constraints.[16]

3 Main result

Fix any function $d^* : \Theta \to D$, which describes all the economically relevant outcomes in this environment except for the information acquisition costs. If the principal could observe θ, then any d^* is attainable without any information acquisition cost on the agents' side. In this sense, one may interpret this d^* together with zero cost as the *first-best* outcome.[17]

In this section, for any given d^*, we explicitly construct a mechanism that implements d^* with zero cost. That is, the first-best outcome is indeed attainable, *even though the principal cannot directly observe θ.*

Theorem 1. *Fix any $d^* : \Theta \to D$. Under Assumption 1, there exists a mechanism (σ, δ) such that (i) $\sum_s \alpha(\theta, s) 1_{\{\delta(s)=d^*(\theta)\}} = \mu_0(\theta)$ for all θ, and (ii) $c_i(\sigma_i) = 0$ for all i.*

Proof. Our construction directly builds on Zhu (2021):

Lemma 1. *(Zhu (2021); Lemma 2)*[18] *Fix any $M, N \in \mathbb{N}$ and any random variable with finite support, $\tau_0 \in \{0, \ldots, K-1\}$. There exists $\tau_1, \ldots, \tau_N$ such that (i) for any $i \in \{1, \ldots, N\}$, (τ_0, τ_i) are pairwise independent, and (ii) for any $i, j, k \in \{1, \ldots, N\}$ with $i \neq j$, $\tau_0|(\tau_i, \tau_j)$ and $\tau_k|(\tau_i, \tau_j)$ are degenerate.*

In our context, let $\tau_0 = \theta$ and $s_i = \tau_i$, and let $\sigma_i \in \Sigma_i$ be the one that induces s_i. Then, the first property says that θ, s_i are pairwise independent, implying $c_i(\sigma_i) = 0$. The second property says that, given s_i, s_j with $i \neq j$, we can fully identify θ and any s_k without error, that is, there exist $\theta(s_i, s_j)$ and $s_k(s_i, s_j)$ such that:

$$\Pr(\theta = \theta(s_i, s_j)|s_i, s_j) = \Pr(s_k = s_k(s_i, s_j)|s_i, s_j) = 1.$$

[16] Ex interim individual rationality is guaranteed because agent i can always deviate to $\tau_i(\cdot) \equiv \phi_i$. Ex ante individual rationality is guaranteed because agent i can always deviate to a costless σ_i and then to $\tau_i(\cdot) \equiv \phi_i$.

[17] For example, one may imagine the principal's own preference and define $d^*(\theta)$ as his best decision given θ.

[18] Zhu (2021) calls the combination of these properties *individually uninformative but aggregately revealing.*

See Zhu (2021) for the detailed construction and proof.

Let the principal recommend the above σ, and offer the decision rule δ as follows: $\delta(s) = d^*(\theta)$ if (i) for any i, j with $i \neq j$, we have

$$\theta = \theta(s_i, s_j);$$

or if (ii) there is i such that, for any j, k where i, j, k are all different, we have

$$\theta = \theta(s_j, s_k).$$

In any other case, $\delta(s)$ is arbitrary.

Clearly, if the agents obey the recommendation and report their signals truthfully, then the first-best outcome is attained. Therefore, we complete the proof by showing that it satisfies incentive compatibility. Take any agent i, and suppose that he deviates to any σ_i' and report $\tau_i(s_i)$ if each s_i is realized. First, his cost of information acquisition increases weakly. Second, his reporting decision does not affect the social decision at all, because the principal executes $\delta(s) = d^*(\theta(s_j, s_k))$ for an arbitrary pair (j, k) which does not include i. Therefore, it is incentive compatible. □

4 Applications

4.1 Full-surplus extraction in common value auctions

Consider the following common value auction environment. The seller (principal) has a single indivisible good, and there are $I \geq 4$ bidders. The value of the good is common to all the bidders, denoted by $\theta \in \Theta$, where Θ is finite. In fact, the analysis of this section can be straightforwardly extended to the case of "non-pure" common values where each i's valuation is $v_i(\theta)$. Let $\mu_0(\theta)$ denote the probability that θ is the bidders' common value.

Each bidder i's payoff is $\theta q_i - t_i - c_i(\sigma_i)$ if he wins the good with probability q_i, pays t_i to the seller, and spends $c_i(\sigma_i)$ as his information acquisition cost. In case he does not participate in the mechanism, his outside-option payoff is 0. The seller's payoff is revenue, $\sum_{i=1}^{I} t_i$.

The first-best expected surplus of this society is the expected common value:

$$\sum_{\theta \in \Theta} \mu_0(\theta)\theta = \mathbb{E}[\theta].$$

There are several cases where the seller can easily earn $\mathbb{E}[\theta]$. First, if the seller *knows* θ, then he can simply post price θ. Even if the seller does not know θ, if the bidders know θ as their common knowledge (i.e., as *free* information), then again the seller can earn $\mathbb{E}[\theta]$. Conversely, if all the bidders are completely *uninformed* (so that each only knows the common prior μ_0), then again, the seller can post price $\mathbb{E}[\theta]$.

Notice that, with costly information acquisition as considered in our paper, neither of the above ideas would work. First, although it might be possible to make every bidder fully learn θ in some equilibrium, it does not yield $\mathbb{E}[\theta]$ as long as full information is strictly costly. Second, if the seller posts price $\mathbb{E}[\theta]$, then each bidder has a strong incentive of knowing whether the true θ is below $\mathbb{E}[\theta]$ or not: If i finds that $\mathbb{E}[\theta|s_i] < \mathbb{E}[\theta]$ given some signal s_i, he would not buy the good. As long as such information is not too costly, the bidder would be better off by acquiring it.

Therefore, with a general information acquisition cost function, the equilibrium information should be somewhere between full and no information, and it is *a priori* unclear how the seller should find the optimal balance of information and rent extraction. Nevertheless, as long as the cost functions satisfy Assumption 1, Theorem 1 implies that the full-surplus extraction is possible.

Corollary 1. *Under Assumption 1, there is a mechanism which yields* $\mathbb{E}[\theta]$ *as the seller's expected revenue (and each bidder earns 0).*

It is worth emphasizing that the logic here is very different from that of Crémer and McLean (1988). In their paper, the seller exploits an *exogeneously given* correlated signal structure, in order to construct a side-bet scheme that extracts the entire surplus. In our case, each bidder can choose any information structure. Indeed, if he prefers, a bidder can choose an information structure such that his information is *independent* from all the other bidders' signals. The Crémer-McLean lottery scheme, therefore, does not work here. Also, in their mechanism, each

bidder's payoff can be strictly negative *ex post*, while in our case, it is zero *ex post*. Indeed, if a mechanism offers a negative ex post payoff in our environment, a bidder would have a strong incentive of getting a signal indicative of that event and then report differently, in order to avoid such a negative payoff.

4.2 First-best implementation in collective decision-making

Consider a committee with a designer (principal) and $I \geq 4$ members (agents) deciding whether to hire or not to hire a job market candidate. Formally, $d \in D = \{h, nh\}$. The quality of the candidate is $\theta \in \Theta$, which is unobserved *ex ante*. The designer and all members of the committee hold a common prior belief $\mu_0 \in \Delta(\Theta)$ about the candidate's quality.

The utility that each member obtains from hiring / not hiring the candidate is defined as follows:

$$u_i(d, \theta) = \begin{cases} u_i(\theta), & \text{if } d = h \\ 0, & \text{if } d = nh \end{cases}$$

Without loss of generality, we assume that $u_i(\theta) = k_i\theta$.

Only the committee members can acquire information about the candidate at cost $c_i(\sigma_i)$. The designer aims to maximize the expected sum of all members' gross utilities.[19] That is, ideally, he wants to hire the candidate if and only if $\sum_i k_i\theta \geq 0$. The first best expected surplus of all committee members is given by:

$$\sum_{\theta | \sum_i k_i\theta \geq 0} \mu_0(\theta) \sum_i k_i\theta = \sum_i k_i \, \mathbb{E}[\theta | \sum_i k_i\theta \geq 0].$$

It is useful to note that the existing literature (see e.g. Li (2001) and Gerardi and Yariv (2008)) typically assumes that the committee members have access to information structures whose realized signals are independently distributed across them, conditional on the state of the world. Under these restrictions, the first best outcome cannot be implemented. There are two main forces that prevent the committee from implementing the first best with these

[19] The result extends to the case where the designer maximizes expected sum of members' net utilities (taking into account the information acquisition costs).

restricted information structures: free-riding problem and conflict of interest. First, when committee members have a conflict of interest, they may prefer not to report their own acquired information truthfully. Second, even if all members share a common preference, information could be underprovided relative to the social optimum, because it is essentially a public good used to make a collective decision. For example, Li (2001) suggests that distorting the decision rule away from the *ex post* optimal rule (which is optimal under exogenous information) could help to alleviate the free-riding issue.

In contrast to the previous literature, our results show that with more flexible (even though still costly) information acquisition the designer can implement the first best outcome. On the one hand, having access to a wider range of information acquisition technologies enlarges the set of feasible deviations for the agents. On the other hand, the principal now has more flexibility in designing information structures recommended to the agents. Given these two opposing effects, it is not immediately clear *a priori* whether the first best outcome becomes more or less difficult to attain. It turns out that the second effect dominates: with a larger set of feasible mechanisms, the principal is able to incentivize the agents to acquire and report their information truthfully, no matter what social choice rule the principal is trying to implement. Indeed, Theorem 1 implies that the first best is implementable as long as Assumption 1 about the cost functions holds.

Corollary 2. *Under Assumption 1, there is a mechanism which yields* $\sum_i k_i \, \mathbb{E}[\theta | \sum_i k_i \theta \geq 0]$ *as the total expected surplus (the decision is made under full information with no cost).*

Our construction helps to resolve both of the issues that prevent first best implementation with conditionally independent signals. Recall that it costs nothing for an agent to acquire an *"individually uninformative"* signal which is assigned to him under the optimal mechanism. Therefore, the distorted provision of information is resolved. Moreover, even if committee members have a conflict of interest, under our mechanism, they cannot do better than being truthful since any unilateral deviation can be detected by the designer.

5 Concluding remarks

It is quite natural that agents may desire to refine their information in response to the mechanism. This paper proposes one possible framework, based on "private-value" cost functions. We show that such a specification leads to an extremely positive result. We conclude the paper with two remarks.

5.1 Interdependent cost functions

One natural criticism may be that our mechanism induces full information if the signals are aggregated, even though any single signal is completely uninformative: would it be reasonable to assume that such σ_i is costless? Because the answer is necessarily yes under Assumption 1, the question is essentially whether Assumption 1 itself is reasonable.

As far as we are aware of, there is no clear consensus nor guidance as to what kinds of correlation structures of the signals are more or less expensive to each agent. There seem to be cases where more positive correlation is more expensive. For example, fix agent 1's private information, and consider agent 2. If acquiring a positively correlated information necessarily means that agent 2 must *steal* (perhaps a part of) agent 1's information, more positive correlation would be more costly. There are, however, opposite situations. Strulovici (2021), for example, considers the environment where hard evidence is scarce in the sense that, if one agent "picks up" a piece of evidence, then it becomes difficult for the others to get the same or similar evidence. There also seem to be cases where *less correlated* signals are more costly: for example, imagine that there exist 2 newspapers, and σ_i corresponds to the decision of which newspapers to buy (including the options of buying both or none). Assume that the information in those newspapers is highly positively correlated, and assume that agent 1 buys (only) newspaper 1 for sure. In this case, agent 2 receives a positively correlated signal regardless of the newspaper he chooses, while in order to obtain something less correlated, he must buy *both* newspapers and orthogonalize newspaper 2's information from newspaper 1's information. Thus, less correlation is more costly.

This discussion suggests that we must be more specific about the microstructure of the

information sources, in order to determine which correlation structures are more costly. Mechanism design with such more specific information acquisition cost structures would certainly be an interesting future direction, and we hope this article could serve as a first step towards that direction.

5.2 Two or three agents

Obviously, the individually-uninformative-but-aggregately-revealing signal structures require four or more agents. With three agents, although it is possible to determine whether *some* agent has unilaterally deviated or not, but it is not possible to identify who is the deviator (and hence to identify the true θ). If there exists $d \in D$ that can serve as a severe punishment for all agents for any given θ, then the same result obtains. See Zhu (2021) too.

With two agents, then each agent has much more freedom. Indeed, the authors work on a separate project with two agents. There, even under Assumption 1, an extremely positive result similar to the one obtained in this paper does not generally hold. The optimal mechanism typically involves some costly information acquisition, and hence, the specification of the cost functions matters even more.

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